

Xiruo Nie

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PROFESSIONAL SUMMARY

Second-year BSc Mathematics and Statistics student at the University of Leeds with experience in Python, R, statistical analysis and data visualisation. Awarded Best of Show and Best Communication and Data Storytelling at ASA DataFest for translating complex data into clear insights.

EDUCATION

BSc Mathematics and Statistics, University of Leeds, Leeds, UK | 2024 - 2027

- Expected Grade: First Class
- Relevant Modules: Statistical Methods, Computational Mathematics and Modelling, Time Series, Optimisation, Linear Algebra, Stochastic Processes, Investigations in Mathematics, Group Theory

EXPERIENCE

GPU Software / Python Intern, MuXi Integrated Circuit Co., Ltd. | Summer 2025

- Assisted in porting cufile-python to the MuXi GPU environment, gaining practical exposure to Python engineering, GPU architecture and high-throughput computing.
- Debugged Python package compatibility issues by tracing dependency-chain errors across software packages, version conflicts and platform-specific environments.
- Applied algorithmic thinking, including trees, heaps, sorting, recursion, shortest paths and hash tables, to strengthen structured problem-solving and implementation skills.
- Explored single-core testing and performance analysis to understand differences across GPU cores and potential routes for runtime optimisation.

PAL Tutor, University of Leeds | 2025.09 - Present

- Supported junior students in mathematics and statistics through peer-assisted learning sessions, explaining technical concepts in a clear and structured way.
- Helped students improve problem-solving methods and revision strategies, developing communication, mentoring and leadership skills.

ACADEMIC & PERSONAL PROJECTS

ASA DataFest, Data Analysis Competition | 2026.03

- Collaborated in a team-based data competition to clean, analyse and visualise a complex real-world dataset under time constraints.
- Conducted exploratory data analysis to identify patterns and translate statistical findings into practical insights for a non-technical judging panel.
- Structured the final presentation around clear data storytelling, combining evidence, visualisation and recommendations to communicate results effectively.

Mathematical Modelling & Numerical Simulation Project — Epidemic Dynamics, University of Leeds | 2026.02-2026.05

- Built a five-compartment epidemic model to represent healthy humans, infected carriers, zombies, quarantined individuals and deceased individuals, translating a real-world style outbreak problem into a structured mathematical framework.
- Applied the forward Euler method to numerically simulate a coupled nonlinear system of differential equations over a long-term simulation horizon.
- Conducted scenario analysis by varying the quarantine rate to compare human extinction, transient coexistence and human survival outcomes.
- Produced graphical analysis and a phase diagram to visualise how the balance between transmission and quarantine rates determines long-term system behaviour.

Time Series Analysis Project, R / Statistical Modelling, University of Leeds

- Analysed historical time series data to identify trends, fit autoregressive models and evaluate residual behaviour using diagnostic testing.
- Communicated statistical results in a report designed for a non-specialist audience, strengthening technical writing and interpretation skills.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Student Leadership, Newsletter and Model United Nations | 2020 - 2025

- Held multiple leadership roles, including Student Society President and Newsletter Lead, coordinating student engagement, content publication at official account and event organisation.
- Participated in youth economic forums and Model United Nations, developing public speaking, cross-cultural communication and policy discussion experience.

AWARDS & ACHIEVEMENTS

- Best of Show, ASA DataFest, 2026
- Best Communication and Data Storytelling, ASA DataFest, 2026

TECHNICAL SKILLS

- Programming / Languages: Python, R, Excel, LaTeX
- Data Analysis: Data cleaning, exploratory data analysis, statistical modelling, simulation, regression, data visualisation
- Algorithmic Thinking: Binary trees, greedy algorithms, heaps / priority queues, union-find, sorting, recursion, shortest paths, two pointers, hash tables
- Communication: Data storytelling, presentation, report writing, peer tutoring, teamwork
- Languages: Mandarin Chinese, English

INTERESTS

- Data storytelling and analytics: interested in turning complex datasets into clear, audience-focused insights and recommendations.
- Quantitative problem-solving: enjoy applying mathematical and statistical reasoning to practical business and technology questions.
- Photography and visual storytelling: interested in using composition and visual clarity to communicate ideas effectively.
- Figure skating and pop dance: demonstrate discipline, consistency and confidence through regular practice and performance.

References available upon request.